

#5: Labce. 20, 38a, 44

Homework 7, Math 3070

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$$4. a) \bar{x} \pm Z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right) = 51.4 \pm 1.96 \left(\frac{2.8}{\sqrt{100}} \right) = \boxed{51.4 \pm 1.96 (.56)}$$

$$b) \bar{x} = 51.4, \sigma = 2.8, n = 100, \alpha = .05$$

$$\bar{x} \pm Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) = 51.4 \pm 1.96 \left(\frac{2.8}{\sqrt{100}} \right) = \boxed{51.4 \pm .5488}$$

$$c) \bar{x} \pm Z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right) = 51.4 \pm 2.576 \left(\frac{2.8}{\sqrt{100}} \right) = \boxed{51.4 \pm .7213}$$

$$e) n = \left(\frac{\sigma Z_{\alpha/2}}{E} \right)^2 = \left(\frac{2.8 \cdot 2.576}{.5} \right)^2 = \boxed{208.0656}$$

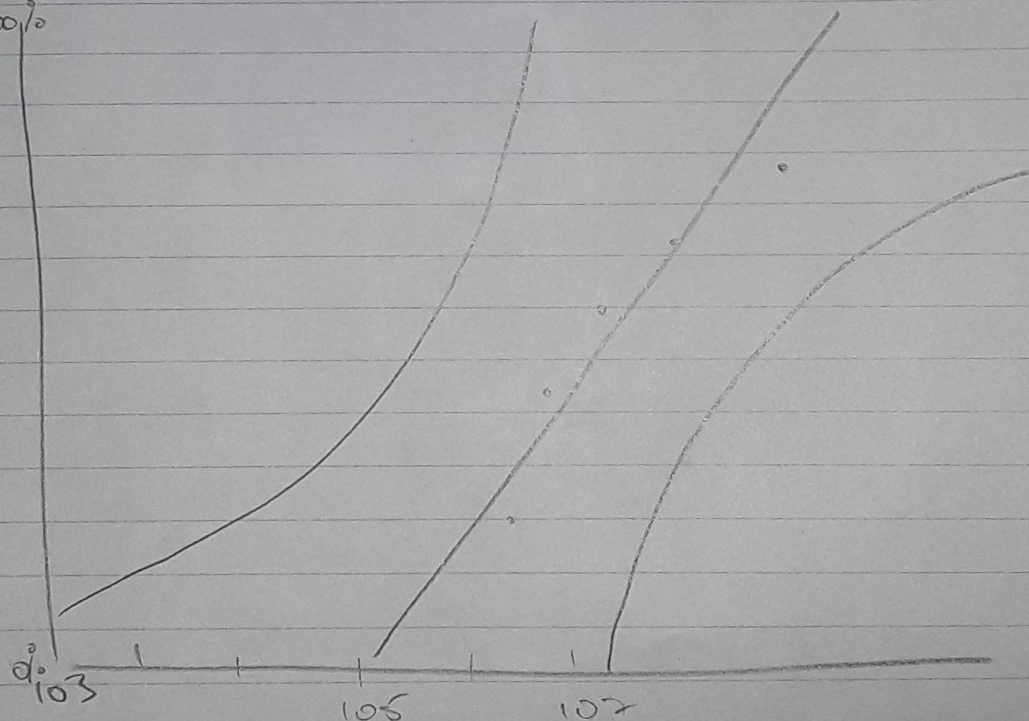
$$20. a) CI = \hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = .53 \pm 2.58 \sqrt{\frac{(.53)(1-.53)}{2343}}$$

$$= \boxed{.53 \pm .027}$$

$$b) n = \frac{4 Z^2 \hat{p}(1-\hat{p})}{w^2} = \frac{4 \cdot 2.58^2 \cdot .53(1-.53)}{(.05)^2}$$

$$= 2652.975 = \boxed{2653}$$

38. a) 100%



44: $n=9$ $\sigma = 2.84$ $CI = 1 - \alpha = .95$ $\alpha = .05$

$$DF = n - 1 = 9 - 1 = 8$$

$$\chi^2_{1-\alpha, n-1} = \chi^2_{0.025, 8} = 17.535$$

$$\chi^2_{1-\alpha, n-1} = \chi^2_{0.975, 8} = 2.180$$

$$\frac{(9-1)(2.84)^2}{17.535} \leq \sigma^2 \leq \frac{(9-1)(2.84)^2}{2.180}$$

$$3.679772 \leq \sigma^2 \leq 29.59853$$

$$\sqrt{3.680} \leq \sigma \leq \sqrt{29.599}$$

$$1.918 \leq \sigma \leq 5.440$$